

DATA ENGINEERING & ANALYTICS

Solving the Last Mile Problem.

A Microsoft Fabric case study in dynamic ingestion routing, metadata-driven control flow, and the operational discipline of active file management for a federated municipal data estate.

City of Metropolis

Municipal data lakehouse · Metropolis Forward initiative
Data Engineering · Orchestration · Governance · Triage

MICROSOFT FABRIC

CASE STUDY 03 · CONTENTS

Sixteen sections. One router. The triage ends.

A municipal data platform team is building a world-class lakehouse on a foundation of manual triage. Files arrive misnamed. Files become digital ghosts. A clerk's typo can take the Mayor's dashboard dark. The fix is one dynamic router and the operational discipline of active file management.

01

The Situation

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SECTION 01 · EXECUTIVE SITUATION

A world-class lakehouse on a foundation of manual triage.

The City of Metropolis is at a critical juncture in its digital transformation. Under the Metropolis Forward initiative, the city has consolidated its disparate data sources into a central cloud-based landing zone. That success has introduced an unforeseen operational crisis: the automated promise of a unified data lakehouse is being throttled by a manual, fragile, and ungoverned triage process.

WHERE THE ANALYTICAL CAPACITY IS GOING

Every day, the city's limited analytical resources are diverted from high-value insights to perform visual inspections of file names and manual execution of ingestion tasks. The team that should be building predictive models for traffic optimisation and crime prevention is instead spending its day reading filenames.

Leadership now faces a defining choice. The current trajectory leads toward data stagnation and technical debt. An automated, intelligent orchestration framework promises to unlock the true potential of the city's data assets.

WHY THE WINDOW IS NARROW

The upcoming fiscal reporting cycle demands a level of data integrity and freshness that the current manual operating model simply cannot provide. Success in this endeavour will enable proactive urban planning and rapid response capabilities.

"Failure ensures continued reliance on incomplete and potentially erroneous information at the exact moment the city is asking its data platform to prove its worth."

ENGINEERING MANDATES

R

Replayability
The architecture must support automatic re-ingestion of historical data without altering the central pipeline logic.

V

Schema Visibility
All ingestion metadata must be observable to governance teams to prevent digital ghost files from going undetected.

S

SLA Observability
Real-time monitoring of ingestion duration and success rates is mandatory for operational trust across departments.

D

Domain Ownership
Data producers must be held accountable for schema compliance via automated validation guardrails at the entry point.

E

Security Enforcement
Least-privilege access must be applied at the storage and compute layers to protect citizen privacy without compromise.

G

Governance by Default
Active file management archives processed files and quarantines unrecognised ones. The landing zone is clean at the end of every cycle.

SECTION 02 · VALUE EXPOSURE

The cost of inaction is no longer theoretical.

Internal audits have identified significant financial and operational leakage stemming from the Last Mile problem. The numbers below quantify the exposure facing the city administration if the ingestion process is not automated. They are conservative estimates. They do not include the political cost of a Mayor's report going dark.

01

Reporting latency cost

Direct labour waste and missed optimisations attributable to the manual triage cycle. Compounds monthly.

\$40K-75K

Per month, ongoing

02

Operational inefficiency

Increase in cross-departmental response times caused by data not being available when decisions need to be made.

20-30%

Cross-departmental slowdown

03

Compliance exposure

Unprocessed sensitive files lingering in the landing zone. Citizen privacy obligations and audit obligations both at risk.

High

Sensitive files unprocessed

04

Data integrity risk

Error rate in manual daily police and transit reports. A defect rate at this level in any other operational system would trigger a halt-and-investigate.

15-25%

Manual report error rate

Compounds across every cross-departmental query that depends on these as source records.

THE STRATEGIC FRAME

Each exposure compounds. The labour cost funds the manual process that creates the integrity defect that erodes the compliance posture. A single automation programme dissolves all four simultaneously because all four trace to the same architectural absence: no automated triage at the entry point. Treating them as separate cost lines is the wrong frame. They are one bill, paid in four currencies.

SECTION 03 · ORGANIZATIONAL CONTEXT

Federated estates, sovereign departments, one central inbox.

The City of Metropolis operates under a federated administrative structure where individual departments have historically maintained complete sovereignty over their data estates. The decentralised model was designed for departmental agility. Under the Metropolis Forward initiative, it has become a primary driver of architectural fragmentation.

WHERE THE FRICTION SITS

The central platform team is enforcing a unified data lakehouse strategy across stakeholders accustomed to siloed operations and non-standardised reporting cycles. The delivery pressure is compounded by a lack of shared technical standards, creating an environment where cross-departmental data correlation is manual and labour-intensive rather than streamlined.

The friction between legacy workflows and modern lakehouse requirements has reached a breaking point.

WHAT THE ANALYTICAL TEAM IS DOING INSTEAD

The central data inbox, intended as a clean entry point, has evolved into a high-variance landing zone. Senior engineers spend significant portions of their day manually validating file headers and correcting naming conventions instead of building predictive models for traffic optimisation or crime prevention.

This operational drag effectively caps the city's analytical throughput and threatens the long-term viability of the digital initiative.

EXHIBIT 01 · ENTERPRISE DATA LANDSCAPE

DEPARTMENT	DOMAIN	TECHNICAL PROFILE	FREQ
Police (MPD)	Incidents	Legacy CSV. Critical PII encryption required. Volume is moderate; sensitivity is highest in the estate.	DAILY
Transport (MDT)	Traffic / Parking	JSON. Unstructured schema with high volatility. Volumes spike around enforcement and events.	DAILY
311 Services	Citizen requests	Structured CSV. High volume text data. The pulse of public sentiment across the city.	DAILY

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We are building a world-class lakehouse on a foundation of manual triage. If a clerk renames a file by mistake, our entire reporting suite for the Mayor goes dark. This is not just a technical debt issue. It is a public accountability crisis that we must automate our way out of immediately.

DIRECTOR OF ANALYTICS · METROPOLIS PLATFORM TEAM

SECTION 04 · CURRENT STATE DIAGNOSIS

Visual matching, digital ghosts, storage bloat.

The diagnosis of the Metropolis landing zone reveals a complete breakdown of automated governance. The current state is defined by fragmented ingestion logic where files are manually identified based on visual name matching. The process is inherently fragile and the symptoms are now measurable.

THE TWELVE PERCENT THAT BREAKS EVERYTHING

A recent audit showed that nearly 12 percent of expected files were either double-loaded or missed entirely due to naming deviations. A file submitted as `Parking_Violations_Final.json` instead of the standard `Parking_Violations.json` is, to the manual process, a different file.

Sometimes it gets loaded twice. Sometimes it gets skipped. Either way the downstream consumer is paying the cost.

WHERE THE DIGITAL GHOSTS COME FROM

The inbox lacks a state-management mechanism. Files are not archived or quarantined upon processing, leading to storage bloat and ambiguity regarding the source of truth for daily records. Unrecognised files become digital ghosts: they occupy space but provide no value as they are never integrated into the analytical layers.

"A file that no system claims responsibility for is a file the city is paying to store and pretending does not exist."

EXHIBIT 02 · CONSTRAINT ANALYSIS

CONSTRAINT	TECHNICAL SYMPTOM	BUSINESS IMPACT
MANUAL TRIAGE	Visual file identification. Engineers read filenames and decide what to do with each one.	High labour cost. Team burnout. Senior analytical capacity absorbed by clerical work.
NO STATE LOGIC	Processed files stay in the inbox. There is no mechanism to mark a file as done.	Risk of data duplication. Audit failure when an external reviewer asks for the source-to-table trail.
HARDCODED PATHS	No dynamic routing. Each source has its own bespoke pipeline definition.	Extensive technical debt. Onboarding a new source requires platform engineering, not a config change.

DIAGNOSTIC INSIGHT

The transition from a messy landing zone to a governed Lakehouse requires more than just compute power. It requires an intelligent orchestration layer that can handle the unpredictability of departmental submissions without manual intervention. Solving the visible problems (the labour cost, the duplicates, the misses) is downstream of solving the structural one: the absence of a router.

SECTION 05 · TARGET OPERATING MODEL

From manual triage to platform-centric orchestration.

The Target Operating Model transitions from a reliance on manual triage to a platform-centric approach where automation is the default. The central platform team functions as an enablement group, providing the structural frameworks and intelligent routing logic for cross-departmental data flow. Departments are designated as domain owners, responsible for the structural integrity and timely delivery of their data assets.

Central platform responsibilities include maintenance of a single, dynamic ingestion framework capable of interpreting incoming file metadata to determine the appropriate processing path. The platform automatically archives successfully processed files to maintain an audit trail and quarantines unrecognised or misnamed files to prevent data loss, effectively eliminating the digital ghost phenomenon. Observability is baked into the ownership model: domain teams monitor the quarantine queues for their specific data domains.

EXHIBIT 03 · OPERATING MODEL SHIFT

FUNCTIONAL AREA	CURRENT STATE (FRAGMENTED)	TARGET STATE (ORCHESTRATED)
Ingestion logic	Manual visual identification and manual triggers. Engineers decide which file goes where.	Dynamic pattern matching via metadata inspection. The pipeline decides.
Error handling	Unrecognised files are ignored. They become digital ghosts in the landing zone.	Automated quarantine and exception queueing. Every file is accounted for somewhere.
Lifecycle management	Files remain in the inbox after processing. No state, no audit trail.	Automatic archival upon successful ingestion. The inbox is empty at end of cycle.
Operational flow	High-touch manual triage process. Daily clerical work.	Low-touch centralised router pipeline. The engineer monitors, not operates.
Domain accountability	Central team absorbs the cost of departmental data quality issues.	Departments own their quarantine queue. Errors return to the source of truth.

ENGINEERING EXPECTATIONS

The framework is built on iteration, conditional logic, and resilient orchestration. New data sources comply with strict onboarding rules including registration of standard naming conventions. The ingestion pattern uses file metadata dynamically. Every cycle produces a verifiable record of which files were processed, archived, or quarantined. Schema contracts are mandatory and structural deviations are identified at the point of entry rather than after the corruption has reached silver or gold.

SECTION 06 · DELIVERY MANDATE

Architect the Single Dynamic Router Pipeline.

You are the Lead Data Engineer reporting to the Enterprise Platform Head for the City of Metropolis. Your primary mission is to architect and implement the Single Dynamic Router Pipeline that will serve as the city's automated triage system. The architecture must provide the optimal balance of power, scalability, and maintainability, avoiding the operational complexity of hardcoded pipelines and the high maintenance overhead of custom code-first solutions.

WHAT THE DESIGN MUST GUARANTEE

01	Active file management	The landing zone remains clean at all times. Successfully processed files are moved to archive; unrecognised files are moved to quarantine. End of cycle, the inbox is empty.
02	Compliance posture	Incident reports from the MPD and sensitive citizen data must be handled with strict adherence to the city's security enforcement mandates. Least-privilege at storage and compute.
03	Full file accounting	The system identifies, routes, and logs 100 percent of files entering the central inbox. Nothing is ignored. Nothing becomes a digital ghost.
04	Auditable analytical backbone	The pipeline produces a robust and auditable analytical backbone for the Metropolis Forward initiative. Source-to-storage lineage must be reconstructable for any file, any day.

THE STRATEGIC FRAME

The mandate is not to build a pipeline. It is to build a piece of platform infrastructure that other engineers, other departments, and future use cases will all depend on without thinking about. The right architecture absorbs new sources without modification. The wrong one is rebuilt every time the city adds a department.

SECTION 07 · DECISION ALTERNATIVES

Three patterns. One recommendation.

The central decision facing the engineering team is not merely technical but strategic: how should the City of Metropolis design a single, intelligent pipeline that acts as an automated triage process for its messy landing zone? The chosen architecture must correctly identify and route known data files, archive them upon success, and quarantine any unrecognised files to ensure a clean, auditable, and reliable ingestion process. Three architectural patterns were evaluated.

Option A Hardcoded

Multiple Hardcoded Pipelines Strengths.

Simple to build. Each pipeline has a narrow, well-understood scope.

Weaknesses.

Massive operational complexity. No central error handling. Every new department requires a new pipeline. Quarantine logic is duplicated across pipelines or omitted entirely.

Verdict: not selected. Scales linearly with departments.

Option B Recommended

Single Dynamic Router Pipeline Strengths.

Centralised routing logic. Scales to new sources without structural changes. Active file management archives processed files and quarantines unrecognised ones. Complete audit trail. One pipeline for the whole city.

Weaknesses.

Requires deeper upfront design knowledge of control flow activities and dynamic expressions such as

```
@item().name
```

.

Recommended. This is what you will build.

Option C Code-First

Code-First Triage via Notebook Strengths.

Maximum flexibility for complex or poorly structured landing zones. Full file system control via PySpark.

Weaknesses.

Introduces a code dependency and specialised skill requirement. Creates key-person risk. Increases total cost of ownership. Moves away from the accessible low-code paradigm.

Verdict: power without proportion. Reserved for genuinely chaotic estates.

EXHIBIT 05 · SOURCE SYSTEM INVENTORY

Three sources. Three filenames. Three priorities.

The router pipeline is designed to recognise three known filename patterns and route each to its appropriate downstream domain in the Bronze layer. Anything that does not match these patterns lands in the quarantine folder, where the originating department is responsible for review.

STANDARD FILENAME PATTERN	SOURCE SYSTEM	INGESTION PRIORITY
Police_Incidents.csv	MPD	Critical · PII Protected
Parking_Violations.json	MDT	High · Urban Planning
311_Service_Requests.csv	311	Medium · Public Sentiment

EXHIBIT 06 · TARGET DATASET REQUIREMENTS

REQUIREMENT	METRIC FOR SUCCESS	ACCOUNTABLE STAKEHOLDER
INBOX CLEANLINESS	Zero files remaining in the landing zone at end of cycle.	Platform Team
AUDIT TRAIL INTEGRITY	100 percent archival correlation. Every processed file accounted for.	Governance Board
EXCEPTION VISIBILITY	Automated quarantine alerting. Anomalies surface within the cycle.	Data Stewards

NOTE ON NAMING DISCIPLINE

The filenames above are the contract. The case for naming standards is not technical; it is operational. Each filename pattern is one IF branch the router can resolve without ambiguity. Every additional name variant a department invents is a branch the system cannot reason about, and therefore a file that ends up in quarantine. The router treats consistency as kindness to the platform.

SECTION 08 · PHASED DELIVERY

Three phases. Foundation first. Scale never first.

The transition from current manual triage to a fully automated Metropolis Data Lakehouse is structured across three strategic phases. This phased approach ensures the foundational orchestration logic is stabilised before scaling to the city's broader data ecosystem.

Phase 01 Foundation

Dynamic Routing & Triage

Replace visual file identification with a dynamic router pipeline that handles the primary datasets from the Police and Transportation departments. This is the proof point. If the router holds for MPD and MDT, it holds for the rest.

Success KPI: zero percent manual daily triage.

Phase 02 Enablement

Domain Onboarding

Formalise the metadata registration process, allowing additional city divisions including Citizen Services to onboard their data via standardised schema contracts. New sources are absorbed by configuration, not engineering.

Success KPI: reduced onboarding time for new departments.

Phase 03 Scale & Stabilise

Observability & Auditing

Implement advanced observability metrics and automated auditing to ensure long-term platform reliability and the elimination of technical debt associated with hardcoded ingestion patterns. The platform becomes self-reporting.

Success KPI: 100 percent audit trail accuracy.

SECTION 09 · ACCEPTANCE CRITERIA

Five benchmarks. Governance sign-off.

To achieve formal sign-off from the City of Metropolis Governance Board, the automated ingestion solution must meet the technical and operational benchmarks below. All five are non-negotiable.

01	Automated Triage	100 percent of files in the landing zone are identified and routed via metadata without manual intervention. The morning clerical sweep no longer exists.
02	Clean Inbox Policy	Successfully processed files are moved to an archive. Unrecognised files are moved to quarantine. The landing zone is empty at the conclusion of every cycle.
03	Auditable Lineage	The system maintains a complete log of file movement from entry to final storage layer. Source-to-table lineage must be reconstructable for any file on any day.
04	Schema Enforcement	The pipeline identifies and rejects files that deviate from the registered department schema. Structural drift is caught at the entry point, not in the gold layer.
05	Operational Resilience	The system handles naming anomalies including <code>_final</code> and <code>_v2</code> suffixes via automated exception logic. No single misnamed file halts the platform.

VALIDATION METHOD

Each criterion is verified through a combination of pipeline monitoring (activity-level success counts, file counts read versus written, exception queue depth) and direct inspection of the landing, archive, and quarantine folders at the end of each scheduled run. The governance board signs off only when all five criteria are demonstrably met across a full week of automated execution.

SECTION 10 · RISK REGISTER

Where this design can still hurt you.

Every automated platform trades one class of risk for another. The router pipeline absorbs the labour cost and the digital ghost problem, but it inherits a new set of failure modes. The list below is what you must be ready to answer for in a governance review.

CATEGORY	RISK DESCRIPTION	MITIGATION STRATEGY
TECHNICAL	Schema drift at the source systems. A departmental vendor changes the structure of an emitted file without notifying the platform team.	Implement automated schema validation at the point of ingestion. Structural deviations are caught before the file is routed and the originating department is alerted via the quarantine queue.
OPERATIONAL	Performance bottlenecks in the Switch logic as the source inventory grows. A router serving fifty departments behaves differently from one serving three.	Optimise iteration concurrency settings. Profile pipeline duration as new sources onboard. Set SLA thresholds that trigger architectural review before performance becomes a user-visible issue.
ADOPTION	Departmental pushback on naming rules. Sovereign domains historically did not need to coordinate filenames with anyone.	Executive mandate for data standards. The router's quarantine behaviour creates an organic enforcement loop: a department that violates the contract sees its file fail to land. Compliance becomes self-policing within one cycle.

THE CULTURAL RISK BEHIND THE TECHNICAL ONE

The adoption risk is the one most likely to derail this. The technical risks have technical mitigations. The adoption risk requires the platform team to hold a line under departmental pressure, and the Director of Analytics to escalate when that line is tested. Naming conventions feel small. They are not. They are the contract that makes the architecture possible.

SECTION 11 · EXECUTIVE DISCUSSION FRAMEWORK

The six questions the City Administrator will ask.

As you present the architectural shift to the City Administrator, prepare to lead a strategic discussion on the following six questions regarding long-term platform viability. None of them are technical. All of them depend on understanding the technical decisions you made.

Key-Person Risk	How does the dynamic router pipeline reduce the key-person risk associated with manual triage? Specifically, what happens if the senior engineer responsible for the morning sweep takes a sabbatical, and how does that compare to the new operating model?
Total Cost of Ownership	What is the total cost of ownership comparison between hardcoded pipelines and this dynamic architecture, projected over a three-year horizon as the city onboards additional departments and data sources?
Quarantine as a Quality Lever	How will the governance team utilise the quarantine folder to improve departmental data quality over time? Is quarantine a punishment, a feedback signal, or both?
No More Graveyard	In what ways does this architecture ensure that the landing zone never becomes a graveyard of unprocessed files? What is the operational discipline that supports the technical guarantee?
Centralised Security Posture	What are the security implications of centralising the ingestion logic for multiple sensitive departments, particularly for PII-bearing sources like MPD incidents? Where do new attack surfaces emerge and how are they mitigated?
Measuring Urban Efficiency	How will we measure the successful realisation of urban efficiency improvements through this automated platform? What is the metric that proves the platform investment paid for itself?

SECTION 12 · LEARNING OBJECTIVES & OUTCOMES

What you will be able to do.

Upon completing this case, you will be able to evaluate, design, and justify an automated ingestion architecture for a multi-source, unstructured landing zone. The objectives below define the scope of that competency. The outcomes define what you will be able to demonstrate.

LEARNING OBJECTIVES

- | | | |
|-----------|-------------------------------------|--|
| 01 | Architectural Trade-Offs | Evaluate the trade-offs between simple hardcoded pipelines and dynamic orchestration frameworks, with specific attention to post-processing file management requirements. |
| 02 | Control Flow Application | Apply fundamental control flow concepts, specifically iteration using a ForEach activity and conditional routing using a Switch activity, to construct an automated triage process that includes routing, archiving, and quarantining. |
| 03 | Resilient Ingestion Patterns | Understand and implement an ingestion pattern that gracefully handles both expected and unexpected files. A critical skill for building robust and auditable data platforms on Microsoft Fabric. |

LEARNING OUTCOMES

A

Architectural Assessment

Confidently assess different architectural patterns for handling ingestion from a multi-source, unstructured landing zone, and justify a chosen approach by articulating its trade-offs against the alternatives.

P

Pipeline Design

Architect and formulate a complete Microsoft Fabric pipeline design that uses file metadata to dynamically control the flow of data and the lifecycle of the source file, from arrival to archival or quarantine.

G

Governance by Design

Design a solution that not only ingests data but also maintains a clean and organised landing zone, ensuring reliability and auditability of the platform through active file management.

SECTION 13 · LAB INSTRUCTIONS

Guided discovery.

Build the router.

You are the Lead Data Engineer on the Metropolis Platform Team. Your task is to construct the Single Dynamic Router pipeline in Microsoft Fabric. The six steps below define the structure of your build. The reasoning behind each step is your portfolio deliverable.

Step 01
Ingest**Retrieve the file list**

Use metadata activities to retrieve a list of all files currently in the landing zone. This list becomes the input to every downstream control flow decision.

Step 02
Audit**Implement success-based archiving**

Configure the archive pattern so that files which complete their routing successfully are moved to an archive location. The inbox is kept clean by construction, not by clean-up jobs running afterwards.

Step 03
Map**Build the Switch activity**

Route files based on name patterns: Police, Transport, 311. Each known pattern matches a Switch case that resolves to the correct downstream domain in the Bronze layer.

Step 04
Verify**Configure the Default path**

The Default branch of the Switch moves unknown files to quarantine. This is the architecture's safety net. Without this branch, an unrecognised file is a digital ghost. With it, every file is accounted for.

Step 05
Schedule**Define the trigger**

Configure the schedule to ensure high-frequency, daily automated execution. The pipeline runs without a human starting it. Operations should monitor, not initiate.

SECTION 14 · EPILOGUE

Two weeks after deployment.

The Single Dynamic Router Pipeline was the recommended and ultimately implemented solution. The architecture was selected because it provided the optimal balance of power, scalability, and maintainability. Upon deployment, the pipeline immediately automated the entire triage process, eliminating hours of daily manual work from the analytics team's schedule.

WHAT THE ACTIVE FILE MANAGEMENT PROVED

The active file management capability proved particularly significant. Successfully processed files were moved to the archive folder, creating a clear and verifiable audit trail. Unrecognised files were moved automatically to the quarantine folder, establishing an exception queue that the data governance team could review on their own cadence.

For the first time in the platform's history, end of cycle meant end of cycle. The inbox was empty. Every file was somewhere.

THE FILE THAT CAME BACK

Two weeks after deployment, the team discovered a misnamed but critical report sitting in the quarantine folder. Under the previous manual process, that file would have been a digital ghost: occupying storage space but providing no analytical value and leaving a gap in the city's reporting suite.

Its detection and recovery validated the new architecture and built significant trust in the automated platform across all participating departments. The quarantine folder was no longer a graveyard. It was a triage hospital.

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The point of the architecture was never to make the platform faster. It was to make it accountable. The first time we pulled a misnamed file out of quarantine and traced its origin back to the originating department within minutes, every objection to the project went quiet. Accountability turned out to be the killer feature.

CLOSING REFLECTION · METROPOLIS PLATFORM TEAM, TWO WEEKS POST GO-LIVE

WHAT THIS CASE ESTABLISHES

A landing zone problem looks like an ingestion problem until you solve it. Then it becomes an accountability problem, then a culture problem, then an enablement problem. The platform team that ships this architecture is not just shipping a pipeline. It is shipping a contract between the city and its data producers, written in the language of metadata.

From visual triage to a router that decides for itself.

Build the pipeline that gives the Metropolis platform team its morning back. Dynamic filename pattern matching, metadata-driven routing, automatic archival on success, automatic quarantine on the default path, complete audit lineage end to end. One pipeline, one schedule, one clean inbox at the close of every cycle.

WHAT YOU WILL DELIVER

A Microsoft Fabric pipeline acting as the city's automated triage system. ForEach iteration over the landing zone. Switch routing on filename pattern. Archive on recognised match. Quarantine on default. Schedule-driven.

HOW YOU WILL BE EVALUATED

Five acceptance criteria covering triage automation, inbox cleanliness, auditable lineage, schema enforcement, and operational resilience. Six executive questions the City Administrator will ask.

WHAT THIS PROVES

You can take a federated organisational structure with sovereign data producers and build a piece of central platform infrastructure that absorbs new sources, enforces standards through quarantine, and stays auditable end to end.

WHAT COMES NEXT

Onboard 311. Add automated quarantine alerts to data stewards. Add schema validation at the entry point. Move the Bronze layer into a wider Silver semantic model. The router is the platform. Everything else is configuration.

INSTRUCTOR CONTACT

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